

Deformacijski tenzor
 $u_{ij} = \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)$
 Napetostni tenzor
 $f_i = \frac{\partial p_{ij}}{\partial x_j}$

Plošče
 $u_z = Z(x, y)$
 $D \Delta^2 Z(x, y) - P(x, y) = 0$
 površinska gost. sil
 $D = \frac{Eh^3}{12(1-\sigma^2)}$

Palice - torzija
 $\varphi = \tau z$
 $\vec{u} = \vec{\varphi} \wedge \vec{r}$
 $\vec{\varphi} = \tau z \hat{e}_z$
 $u_x = -\tau y z$
 $u_y = \tau x z$
 $u_z = \tau \varphi(x, y)$
 $u_{xz} = \frac{1}{2} (-\tau y + \tau \frac{\partial \varphi}{\partial x})$
 $u_{yz} = \frac{1}{2} (\tau x + \tau \frac{\partial \varphi}{\partial y})$
 $p_{xz} = \mu \tau (-y + \frac{\partial \varphi}{\partial x})$
 $p_{yz} = \mu \tau (x + \frac{\partial \varphi}{\partial y})$
 ostali u_{ij} in p_{ij} so 0
 Ravnovesje:
 $\nabla^2 \varphi = 0$
 $f_i = \frac{\partial p_{ij}}{\partial x_j} = 0 \rightarrow f_z = \frac{\partial p_{zx}}{\partial x} + \frac{\partial p_{zy}}{\partial y} = 0$
 Privrejena torzijska funkcija:
 $p_{xz} = 2\mu \tau \frac{\partial \chi}{\partial y}$
 $p_{yz} = 2\mu \tau \frac{\partial \chi}{\partial x}$
 $\frac{\partial \varphi}{\partial x} = y + 2 \frac{\partial \chi}{\partial y}$
 $\frac{\partial \varphi}{\partial y} = -x - 2 \frac{\partial \chi}{\partial x}$

Cauchyjev izraz
 $\sigma_{ij} = f_i + \frac{\partial p_{ik}}{\partial x_k}$
 ↑
 zun. sile
 ravnovesje:
 $f_i + \frac{\partial p_{ik}}{\partial x_k} = 0$
 $\rightarrow p_{ik} n_k = \vec{F}_i$
 ↑
 povr. sile
 zun. normala

RP:
 1) togo upeta plošča
 $Z|_{rob} = 0$
 $\frac{\partial Z}{\partial n}|_{rob} = 0$
 2) priložena plošča
 $Z|_{rob} = 0$
 $(\frac{\partial^2 Z}{\partial n^2} + \sigma \frac{\partial g}{\partial c} \frac{\partial Z}{\partial n})|_{rob} = 0$
 - krog: $\frac{\partial y}{\partial c} = \frac{1}{R}$

Hookov zakon
 $p_{ij} = 2\mu u_{ij} + \lambda u_{kk} \delta_{ij}$
 $= \frac{E}{1+\sigma} (u_{ij} + \frac{\sigma}{1-2\sigma} u_{kk} \delta_{ij})$
 $u_{ij} = \frac{1}{2\mu} p_{ij} - \frac{\lambda}{2\mu(2\mu+3\lambda)} p_{kk} \delta_{ij}$
 $= \frac{1}{E} [(1-\sigma) p_{ij} - \sigma p_{kk} \delta_{ij}]$

RP:
 $p_{zx} n_x + p_{zy} n_y = 0$
 $\chi|_{rob} = \text{konst.} = 0$
 $\nabla^2 \chi = -1$
 $M_z = -2\mu \tau \int dS [x \frac{\partial \chi}{\partial x} + y \frac{\partial \chi}{\partial y}] = 4\mu \tau \int \chi dS$
 enostavno povezano območje

Navierova enačba
 $\sigma_{ij} = f_i + \mu \nabla^2 u_i + (\mu + \lambda) \frac{\partial}{\partial x_i} (\frac{\partial u_k}{\partial x_k})$
 $\sigma_{ij} = \vec{f} + \mu \nabla^2 \vec{u} + (\mu + \lambda) \nabla (\nabla \cdot \vec{u})$

Palice - upogib
 $\vec{M} = EI (-y'', x'', \varphi)$
 $F_x = -EI x^{(3)} + F_2' x'$
 $F_y = -EI y^{(3)} + F_2' y'$
 ravnovesje:
 $EI x^{(4)} - F_2 x'' - F_2' x' - k_x = 0$
 $EI y^{(4)} - F_2 y'' - F_2' y' - k_y = 0$
 ↓
 dolžinska gost. sile

$\frac{\Delta V}{V} = \text{tr } \underline{u}$
 $p_{ij} = -p$

RP:
 1) Togo upeta
 $x|_{rob} = 0$
 $x'|_{rob} = 0$
 2) prost
 $\vec{F}|_{rob} = 0$
 $\vec{M}|_{rob} = 0$
 3) podprta / vrtljivo upeta
 $x|_{rob} = 0$
 $\vec{M}|_{rob} = 0$

Kristali
 $f(\underline{u}) = \frac{1}{2} C_{ijkl} u_{ij} u_{kl} (+f_0)$
 $p_{ij} = \frac{\partial f}{\partial u_{ij}} = C_{ijkl} u_{kl}$
 $u_{ij} = u_{ji}$
 $C_{ijkl} = C_{jikl} = C_{jilk} = C_{ijlk}$
 $\rightarrow \text{max 21 neodv.}$
 - upoštevaš simetrije

Elastična valovanja
 $C_T^2 = \frac{E}{2\rho(1+\sigma)}$
 $C_L^2 = \frac{E(1-\sigma)}{\rho(1+\sigma)(1-2\sigma)}$
 $\frac{C_T^2}{C_L^2} = \frac{1-2\sigma}{2(1-\sigma)} \leq \frac{1}{2}$
 $\omega = ck$
 $p_{ij} = 2\rho C_T^2 u_{ij} + \rho (C_L^2 - 2C_T^2) u_{kk} \delta_{ij}$
 specifična impedanca
 $Z = \frac{\rho}{v}$
 Navierova enačba 2
 $\frac{E}{2(1+\sigma)} \left[\frac{2(1-\sigma)}{1-2\sigma} \nabla \nabla \cdot \vec{u} - \nabla \times \nabla \times \vec{u} \right] + \vec{f} = \rho \ddot{\vec{u}}$

Sferne Besselove funkcije

$$\frac{d^2}{dx^2} z(x) + \frac{2}{x} \frac{dz(x)}{dx} + \left(1 - \frac{l(l+1)}{x^2}\right) z(x) = 0$$

→ j_l, n_l

Besselove funkcije

$$x^2 \frac{d^2 z}{dx^2} + x \frac{dz}{dx} + (x^2 - \alpha^2) z = 0$$

→ J_α, N_α

Nabla u sfernih koordinatah

$$\nabla f = \frac{\partial f}{\partial r} \hat{e}_r + \frac{1}{r} \frac{\partial f}{\partial \theta} \hat{e}_\theta + \frac{1}{r \sin \theta} \frac{\partial f}{\partial \varphi} \hat{e}_\varphi$$

$$\nabla^2 f = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial f}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial f}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 f}{\partial \varphi^2}$$

$$\nabla \cdot \vec{A} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 A_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta A_\theta) + \frac{1}{r \sin \theta} \frac{\partial A_\varphi}{\partial \varphi}$$

$$\nabla \times \vec{A} = \frac{1}{r \sin \theta} \left[\frac{\partial}{\partial \theta} (A_\varphi \sin \theta) - \frac{\partial A_\theta}{\partial \varphi} \right] \hat{e}_r + \frac{1}{r} \left[\frac{1}{\sin \theta} \frac{\partial A_r}{\partial \varphi} - \frac{\partial}{\partial r} (r A_\varphi) \right] \hat{e}_\theta + \frac{1}{r} \left[\frac{\partial}{\partial r} (r A_\theta) - \frac{\partial A_r}{\partial \theta} \right] \hat{e}_\varphi$$

Nabla u kartezijnih koordinatah

$$\nabla f = \frac{\partial f}{\partial x} \hat{e}_x + \frac{\partial f}{\partial y} \hat{e}_y + \frac{\partial f}{\partial z} \hat{e}_z$$

$$\nabla \cdot \vec{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

$$\nabla \times \vec{A} = \left[\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right] \hat{e}_x + \left[\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right] \hat{e}_y + \left[\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right] \hat{e}_z$$

Nabla u cil. koord.

$$\nabla f = \frac{\partial f}{\partial \rho} \hat{e}_\rho + \frac{1}{\rho} \frac{\partial f}{\partial \varphi} \hat{e}_\varphi + \frac{\partial f}{\partial z} \hat{e}_z$$

$$\nabla^2 f = \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial f}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 f}{\partial \varphi^2} + \frac{\partial^2 f}{\partial z^2}$$

$$\nabla \cdot \vec{A} = \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho A_\rho) + \frac{1}{\rho} \frac{\partial A_\varphi}{\partial \varphi} + \frac{\partial A_z}{\partial z}$$

$$\nabla \times \vec{A} = \left(\frac{1}{\rho} \frac{\partial A_z}{\partial \varphi} - \frac{\partial A_\varphi}{\partial z} \right) \hat{e}_\rho + \left(\frac{\partial A_\rho}{\partial z} - \frac{\partial A_z}{\partial \rho} \right) \hat{e}_\varphi + \frac{1}{\rho} \left(\frac{\partial}{\partial \rho} (\rho A_\varphi) - \frac{\partial A_\rho}{\partial \varphi} \right) \hat{e}_z$$

$$\nabla \times (\nabla f) = 0$$

$$\nabla \cdot (\nabla \times \vec{v}) = 0$$

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}$$

$$\nabla(fg) = f \nabla g + g \nabla f$$

$$\nabla(\vec{u} \cdot \vec{v}) = \vec{u} \times (\nabla \times \vec{v}) + \vec{v} \times (\nabla \times \vec{u}) + (\vec{u} \cdot \nabla) \vec{v} + (\vec{v} \cdot \nabla) \vec{u}$$

$$\nabla \cdot (f \vec{u}) = f(\nabla \cdot \vec{u}) + \vec{u} \cdot (\nabla f)$$

$$\nabla \cdot (\vec{u} \times \vec{v}) = \vec{v} \cdot (\nabla \times \vec{u}) - \vec{u} \cdot (\nabla \times \vec{v})$$

$$\nabla \times (f \vec{v}) = (\nabla f) \times \vec{v} + f(\nabla \times \vec{v})$$

$$\nabla \times (\vec{u} \times \vec{v}) = \vec{u}(\nabla \cdot \vec{v}) - \vec{v}(\nabla \cdot \vec{u}) + (\vec{v} \cdot \nabla) \vec{u} - (\vec{u} \cdot \nabla) \vec{v}$$

$$\nabla^2 \phi = -\rho(\vec{r})$$

$$\rightarrow \phi(\vec{r}) = \int \frac{\rho(\vec{r}')}{4\pi |\vec{r} - \vec{r}'|} d^3 r'$$

$$\nabla^2 \phi = \nabla \cdot (\nabla \phi) = \nabla \cdot \left(\frac{1}{r^2} \nabla \phi \right) + (\vec{v} \cdot \nabla) \vec{v}$$

Def. tenzor u cil. koord.

$$u_{rr} = \frac{\partial u_r}{\partial r}$$

$$u_{\varphi\varphi} = \frac{\partial u_\varphi}{\partial \varphi} + \frac{u_r}{r}$$

$$u_{zz} = \frac{\partial u_z}{\partial z}$$

$$u_{r\varphi} = \frac{1}{2} \left(\frac{\partial u_\varphi}{\partial r} - \frac{u_\varphi}{r} + \frac{\partial u_r}{\partial \varphi} \right)$$

$$u_{rz} = \frac{1}{2} \left(\frac{\partial u_r}{\partial z} + \frac{\partial u_z}{\partial r} \right)$$

$$u_{\varphi z} = \frac{1}{2} \left(\frac{\partial u_z}{\partial \varphi} + \frac{\partial u_\varphi}{\partial z} \right)$$

Def. tenzor u sfericnim koord.

$$u_{rr} = \frac{\partial u_r}{\partial r}$$

$$u_{\theta\theta} = \frac{\partial u_\theta}{\partial \theta} + \frac{u_r}{r}$$

$$u_{\varphi\varphi} = \frac{1}{\sin \theta} \frac{\partial u_\varphi}{\partial \varphi} + \frac{u_r}{r} + \frac{u_\theta}{r \tan \theta}$$

$$u_{r\theta} = \frac{1}{2} \left(\frac{\partial u_\theta}{\partial r} - \frac{u_\theta}{r} + \frac{\partial u_r}{\partial \theta} \right)$$

$$u_{r\varphi} = \frac{1}{2} \left(\frac{\partial u_\varphi}{\partial r} - \frac{u_\varphi}{r} + \frac{1}{\sin \theta} \frac{\partial u_r}{\partial \varphi} \right)$$

$$u_{\theta\varphi} = \frac{1}{2} \left(\frac{\partial u_\varphi}{\partial \theta} - \frac{u_\varphi}{r \tan \theta} + \frac{1}{\sin \theta} \frac{\partial u_\theta}{\partial \varphi} \right)$$

Vzt. momenti

točkasto telc $m r^2$

plazce valja $m r^2$

valj $\frac{1}{2} m r^2$

palica $\frac{1}{12} m l^2$ (u sredini)

krogla $\frac{2}{5} m r^2$

ustla krogla $\frac{2}{5} m \frac{R^2 - r^2}{R^2 + r^2}$

torus $m(R^2 + \frac{3}{2} r^2)$

kvadar $\frac{1}{12} m(a^2 + b^2)$